

Zero Day Initiative — CVE-2019-0604: Details of a Microsoft SharePoint RCE Vulnerability

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Content

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Blog

CVE-2019-0604: Details of a Microsoft SharePoint RCE Vulnerability

** March 13, 2019

** Guest Blogger

*Last month, Microsoft released patches to address two remote code execution (RCE) vulnerabilities in SharePoint. In both Critical-rated cases, an attacker could send a specially crafted request to execute their code in the context of the SharePoint application pool and the SharePoint server farm account. Both of these bugs were reported to the ZDI program by ****Markus Wulftange***. He has graciously provided the following write-up on the details of [CVE-2019-0604](#). **

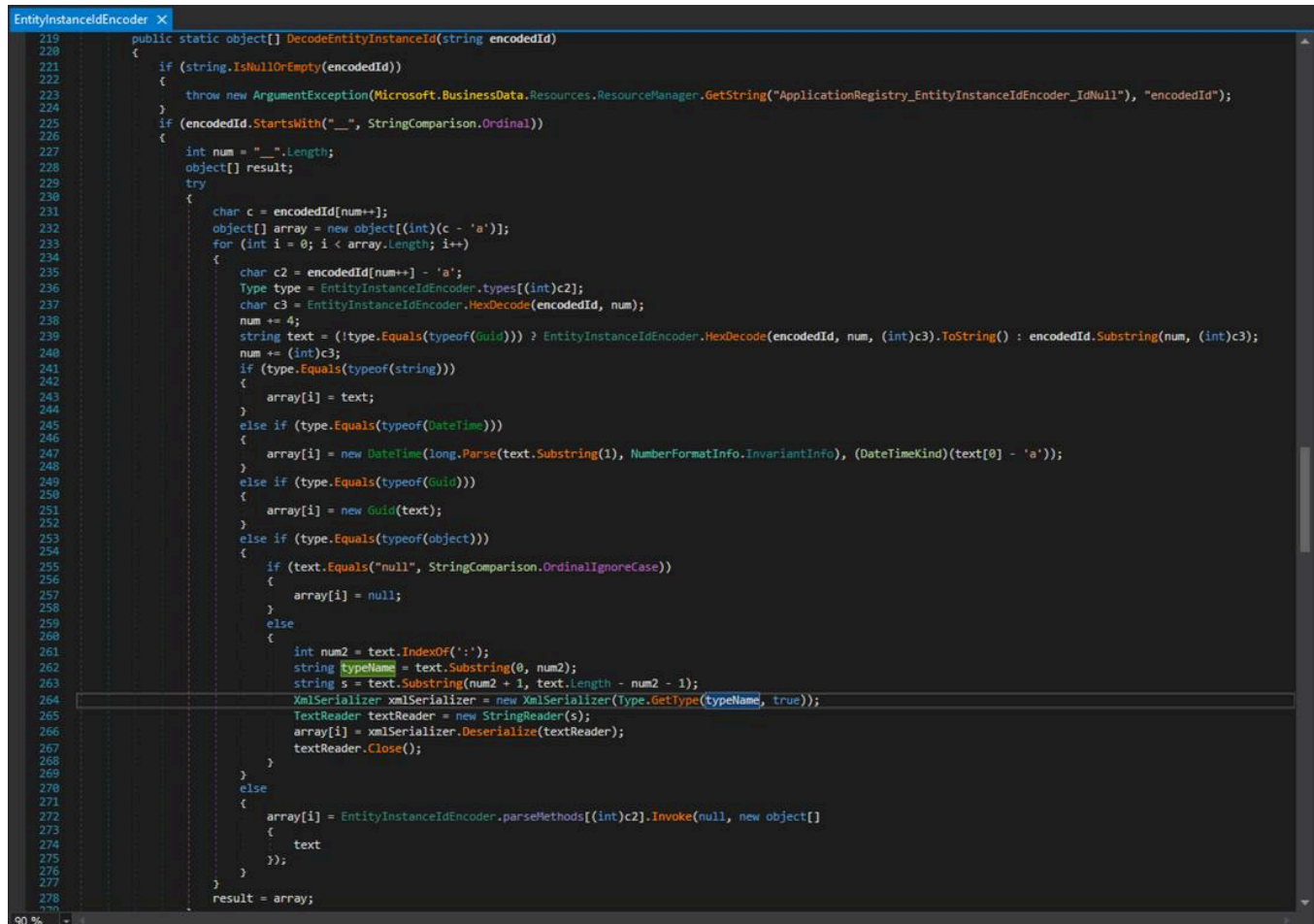
When searching for new vulnerabilities, one approach is the bottom-up approach. It describes the approach of looking for an interesting sink and tracing the control and data flow backwards to find out if the sink can be reached.

One of these promising sinks is the deserialization using the *XmlSerializer*. In general, it is considered a secure serializer as it must be instrumented with the expected type and it is not possible to specify an arbitrary type within the stream that cannot appear in the object graph of the expected type. But it is exploitable if the expected type can be controlled as well, as it has been shown in [Friday the 13th – JSON Attacks by Alvaro Muñoz & Oleksandr Mirosh](#) [PDF].

For analyzing the SharePoint 2016 assemblies, [dnSpy](#) is an excellent tool as it can be used for both decompiling and debugging of .NET applications. So, after dnSpy is attached to the IIS worker process w3wp.exe that is running SharePoint 2016, and the assemblies have been loaded, the usage of the `XmlSerializer(Type)` constructor can be analyzed. Now the tedious part begins where every one of the `XmlSerializer(Type)` constructor calls has to be looked at and to check whether the expected type is variable at all (e.g. it is not hard-coded as in `new XmlSerializer(typeof(DummyType))`) and whether it is possible to control the type.

One of the methods where the `XmlSerializer(Type)` constructor gets called is the `Microsoft.SharePoint.BusinessData.Infrastructure.EntityInstanceIdEncoder.DecodeEntityInstanceId(string)` method in `Microsoft.SharePoint.dll`. The same type with the same functionality is also in the `Microsoft.Office.Server.ApplicationRegistry.Infrastructure` namespace in the `Microsoft.SharePoint.Portal.dll`. We will come back to this later and stick to the one in `Microsoft.SharePoint.dll`.

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```
EntityInstanceIdEncoder X
219 public static object[] DecodeEntityInstanceId(string encodedId)
220 {
221     if (string.IsNullOrEmpty(encodedId))
222     {
223         throw new ArgumentException(Microsoft.BusinessData.Resources.ResourceManager.GetString("ApplicationRegistry_EntityInstanceIdEncoder_IdNull", "encodedId"));
224     }
225     if (encodedId.StartsWith("__", StringComparison.Ordinal))
226     {
227         int num = "__".Length;
228         object[] result;
229         try
230         {
231             char c = encodedId[num++];
232             object[] array = new object[(int)(c - 'a')];
233             for (int i = 0; i < array.Length; i++)
234             {
235                 char c2 = encodedId[num++] - 'a';
236                 Type type = EntityInstanceIdEncoder.types[(int)c2];
237                 char c3 = EntityInstanceIdEncoder.HexDecode(encodedId, num);
238                 num += 4;
239                 string text = (type.Equals(typeof(Guid)) ? EntityInstanceIdEncoder.HexDecode(encodedId, num, (int)c3).ToString() : encodedId.Substring(num, (int)c3);
240                 num += (int)c3;
241                 if (type.Equals(typeof(string)))
242                 {
243                     array[i] = text;
244                 }
245                 else if (type.Equals(typeof(DateTime)))
246                 {
247                     array[i] = new DateTime(long.Parse(text.Substring(1), NumberFormatInfo.InvariantInfo), (DateTimeKind)(text[0] - 'a'));
248                 }
249                 else if (type.Equals(typeof(Guid)))
250                 {
251                     array[i] = new Guid(text);
252                 }
253                 else if (type.Equals(typeof(object)))
254                 {
255                     if (text.Equals("null", StringComparison.OrdinalIgnoreCase))
256                     {
257                         array[i] = null;
258                     }
259                     else
260                     {
261                         int num2 = text.IndexOf(':');
262                         string typeName = text.Substring(0, num2);
263                         string s = text.Substring(num2 + 1, text.Length - num2 - 1);
264                         XmlSerializer xmlSerializer = new XmlSerializer(Type.GetType(typeName, true));
265                         TextReader textReader = new StringReader(s);
266                         array[i] = xmlSerializer.Deserialize(textReader);
267                         textReader.Close();
268                     }
269                 }
270                 else
271                 {
272                     array[i] = EntityInstanceIdEncoder.parseMethods[(int)c2].Invoke(null, new object[]
273                     {
274                         text
275                     });
276                 }
277             }
278             result = array;
279         }
280     }
281 }
```

*

*Figure 1 :

*Microsoft.SharePoint.BusinessData.Infrastructure.EntityInstanceIdEncoder.DecodeEntityInstance(string)

*

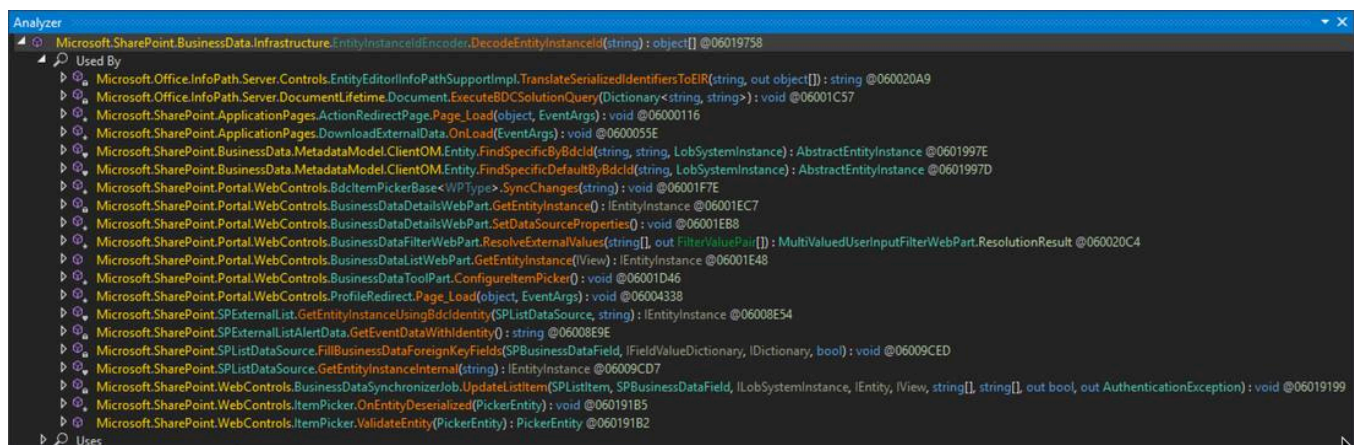
Here both the `typeName` , used to specify the expected type, and the data that gets deserialized originate from `text` , which originates from the method's argument `encodedId` .

This looks perfect as long as the method gets actually called and the passed parameter can be controlled.

Tracing back the Flow to the Source

The next step is to go through the calls and see if the one of them originates from a point that can be initiated from outside and whether the argument value can also be supplied.

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*

*Figure 2: Calls to

*Microsoft.SharePoint.BusinessData.Infrastructure.EntityInstanceIdEncoder.DecodeEntityInstance(string)

*

If you're familiar with the ASP.NET, some of the methods might look familiar like `Page_Load(object, EventArgs)` or `OnLoad(EventArgs)` . They are called during the [ASP.NET life cycle](#), and the types they are defined in extend `System.Web.UI.Page`, the base type that represents `.aspx` files. And, in fact, all three types have a corresponding `.aspx` file:

· *Microsoft.SharePoint.ApplicationPages.ActionRedirectPage:*

`/_layouts/15/ActionRedirect.aspx`

· *Microsoft.SharePoint.ApplicationPages.DownloadExternalData:*

`/_layouts/15/downloadexternaldata.aspx`

· *Microsoft.SharePoint.Portal.WebControls.ProfileRedirect:*

`/_layouts/15/TenantProfileAdmin/profileredirect.aspx`

Although in all three cases the parameter value originates from the HTTP request, it is from the URL's query string. That might become a problem as the hex encoding will multiply the length by 4 and thereby can get pretty long and exceed the limit of the HTTP request line.

After further analysis, the last one of all, the `ItemPicker.ValidateEntity(PickerEntity)` method, turned out to be a better pick.

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```
378 // Token: 0x06019182 RID: 102834 RVA: 0x008EA454 File Offset: 0x008E8654
379 public override PickerEntity ValidateEntity(PickerEntity pe)
380 {
381     bool isSpecificInstance = false;
382     bool flag = false;
383     AuthenticationException bdcAuthenticationException = null;
384     if (pe == null)
385     {
386         throw new ArgumentNullException("pe");
387     }
388     IEntityInstance entityInstance = null;
389     this.CurrentMatches = new List<IEntityInstance>();
390     if (!string.IsNullOrEmpty(pe.Key))
391     {
392         try
393         {
394             if (Microsoft.SharePoint.BusinessData.Infrastructure.EntityInstanceIdEncoder.IsEncodedIdentifier(pe.Key))
395             {
396                 object[] identifierValues = Microsoft.SharePoint.BusinessData.Infrastructure.EntityInstanceIdEncoder.DecodeEntityInstanceId(pe.Key);
397                 entityInstance = this.BdcEntity.FindSpecific(new Identity(identifierValues), this.SystemInstance);
398                 isSpecificInstance = true;
399                 this.CurrentMatches.Add(entityInstance);
400             }
401         }
402     }
```

*

*Figure 3: *ItemPicker.ValidateEntity(PickerEntity)

*

Here, the `PickerEntity`'s `Key` property of the passed `PickerEntity` is used in the `EntityInstanceIdEncoder.DecodeEntityInstanceId(string)` call. It gets called by `EntityEditor.Validate()`, which iterates each entry stored in the `EntityEditor.Entities` property to validate it.

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```
588 [SharePointPermission(SecurityAction.Demand, ObjectModel = true)]
589 public virtual void Validate()
590 {
591     if (this.m_Validated)
592     {
593         return;
594     }
595     this.IsValid = true;
596     this.m_entityHash = this.m_entityHashTemp;
597     this.m_listOrder = this.m_listOrderTemp;
598     this.m_Validated = true;
599     if (!this.Visible)
600     {
601         return;
602     }
603     if (this.MaximumEntities > 0 && this.m_listOrder.Count > this.MaximumEntities)
604     {
605         this.IsValid = false;
606         this.ErrorMessage = SPResource.GetString("PickerSelectTooMany", new object[]
607         {
608             this.MaximumEntities
609         });
610         return;
611     }
612     Hashtable hashtable = new Hashtable();
613     ArrayList arrayList = new ArrayList();
614     StringBuilder stringBuilder = new StringBuilder();
615     foreach (object obj in this.Entities)
616     {
617         PickerEntity pickerEntity = (PickerEntity)obj;
618         PickerEntity pickerEntity2 = pickerEntity;
619         try
620         {
621             using (new SPMonitoredScope("Validate a person or entity", 250u, new ISPScopePerformanceMonitor[0]))
622             {
623                 pickerEntity2 = this.ValidateEntity(pickerEntity);
```

*

*Figure 4: *EntityEditor.Validate()

*

That method gets called by `EntityEditor.LoadPostData(string, NameValueCollection)`, which implements the `System.Web.UI.IPostBackDataHandler.LoadPostData(string, NameValueCollection)` method.

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```
702 [SharePointPermission(SecurityAction.Demand, ObjectModel = true)]
703 public virtual bool LoadPostData(string postDataKey, NameValueCollection values)
704 {
705     this.EnsureChildControls();
706     string text = EntityEditor.StrEatUpNbsp(this.HiddenSpanData.Value);
707     this.IsChanged = this.ParseSpanData(text);
708     if (!string.IsNullOrEmpty(text))
709     {
710         foreach (object obj in this._listOrderTemp)
711         {
712             PickerEntity pickerEntity = (PickerEntity)obj;
713             if (pickerEntity.IsResolved)
714             {
715                 this._listRevalidation.Add(pickerEntity);
716             }
717             pickerEntity.IsResolved = false;
718         }
719         this.Validate();

```

*

*Figure 5: *EntityEditor.LoadPostData(string, NameValueCollection)

*

So that method gets automatically called on post back requests to `ItemPicker` web controls. The call graph looks as follows:

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```
Microsoft.SharePoint.BusinessData.Infrastructure.EntityInstanceIdEncoder.DecodeEntityInstanceId(string) : object[] @06019753
    Used By

    Microsoft.SharePoint.WebControls.ItemPicker.ValidateEntity(PickerEntity) :
PickerEntity @060191AD
    Used By

    Microsoft.SharePoint.WebControls.EntityEditor.Validate() : void @06006457
    Used By

    Microsoft.SharePoint.WebControls.EntityEditor.LoadPostData(string,
NameValueCollection) : bool @0600645B

```

Also note the type hierarchy:

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[\[image: image\]](#)

Verifying the Data Flow

Now that there is a way to reach the

`EntityInstanceIdEncoder.DecodeEntityInstanceId(string)` from an `ItemPicker` web control post back, it is still unclear whether the `Key` property of a `PickerEntity` can be controlled as well.

The `EntityEditor.Entities` property is backed by the private field `m_listOrder`, which gets only assigned at two points: during instantiation and within the `EntityEditor.Validate()` method. In the latter case, it gets the value of the private field `m_listOrderTemp` assigned (see line 597 in Fig. 4 above). That field, again, also only gets assigned at two points:

during instantiation and within the `EntityEditor.ParseSpanData(string)` method. This method is also called by `EntityEditor.LoadPostData(string, NameValueCollection)` with the value of an `HtmlInputHidden` and the name "hiddenSpanData" (see line 707 in Fig. 5 above). That field's value can be controlled by the user.

What is left is to see what `EntityEditor.ParseSpanData(string)` does with the passed data and whether it ends up as a `PickerEntity`'s `Key`. We'll skip that because `EntityEditor.ParseSpanData(string)` is pretty long to show and unless it contains special constructs of nested `<SPAN>` and `<DIV>` tags, which get parsed out, everything else ends up in the `PickerEntity`'s `Key` and then in `m_listOrderTemp` list.

So, now we've found and traversed a vector that allows us to reach `EntityInstanceIdEncoder.DecodeEntityInstanceId(string)` from an `ItemPicker`'s post back handling while also having control over the input. What is still left is to find an instance of that web control.

Finding the Entry Point

The `ItemPicker` web control is actually never used directly in an `.aspx` page. But when looking at the usages of its base type, `EntityEditorWithPicker`, it turned out that there is a `Picker.aspx` at `/_layouts/15/Picker.aspx` that uses it – what a coincidence!

That page expects the type of the picker dialog to use to be provided via the "PickerDialogType" URL parameter in the form of its assembly-qualified name. Here, any of the two `ItemPickerDialog` types can be used:

- `Microsoft.SharePoint.WebControls.ItemPickerDialog` in *Microsoft.SharePoint.dll*
- `Microsoft.SharePoint.Portal.WebControls.ItemPickerDialog` in *Microsoft.SharePoint.Portal.dll*

Using the first `ItemPickerDialog` type shows the following page:

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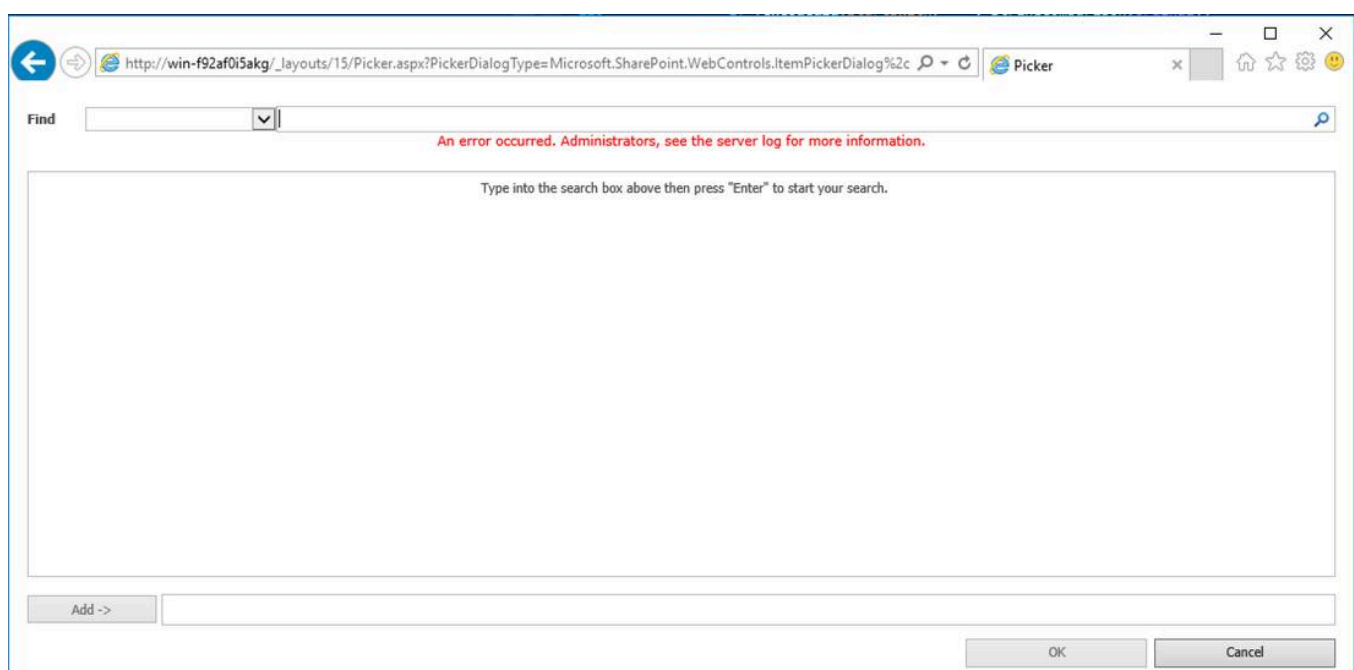


Figure 6: Picker.aspx with Microsoft.SharePoint.WebControls.ItemPickerDialog

Here, the bottom text field is associated to the `ItemPicker`. And there is also the correspondent of the `HtmlInputHidden` with the name `ctl00$PlaceholderDialogBodySection$ctl05$hiddenSpanData` that we were looking for. This is the source of our `EntityInstanceIdEncoder.DecodeEntityInstanceId(string)` sink.

When the form gets submitted with a

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*

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```

>
Microsoft.SharePoint.dll!Microsoft.SharePoint.BusinessData.Infrastructure.EntityInstanceIdEncoder.DecodeEntityInstanceId(string encodedId) (IL=0x0000, Native=0x00007FF89EA38790+0x29)
Microsoft.SharePoint.dll!Microsoft.SharePoint.WebControls.ItemPicker.ValidateEntity(Microsoft.SharePoint.WebControls.PickerEntity pe) (IL=0x003E, Native=0x00007FF89EA361B0+0xD7)
Microsoft.SharePoint.dll!Microsoft.SharePoint.WebControls.EntityEditor.Validate() (IL=0x00D0, Native=0x00007FF89E990740+0x2A3)
Microsoft.SharePoint.dll!Microsoft.SharePoint.WebControls.EntityEditorWithPicker.Validate() (IL=0x001B, Native=0x00007FF89E98FF10+0x7D)
Microsoft.SharePoint.dll!Microsoft.SharePoint.WebControls.ItemPicker.Validate() (IL=0x0006, Native=0x00007FF89EA36010+0x2C)
Microsoft.SharePoint.dll!Microsoft.SharePoint.WebControls.EntityEditor.LoadPostData(string postDataKey, System.Collections.Specialized.NameValueCollection values) (IL=0x008D, Native=0x00007FF89E06A410+0x165)
System.Web.dll!System.Web.UI.Page.ProcessPostData(System.Collections.Specialized.NameValueCollection postData, bool fBeforeLoad) (IL=0x0193, Native=0x00007FF89D075CD0+0x3A1)
System.Web.dll!System.Web.UI.Page.ProcessRequestMain(bool includeStagesBeforeAsyncPoint, bool includeStagesAfterAsyncPoint) (IL=0x0530, Native=0x00007FF89A483B20+0x1024)
System.Web.dll!System.Web.UI.Page.ProcessRequest(bool includeStagesBeforeAsyncPoint, bool includeStagesAfterAsyncPoint) (IL=0x003C, Native=0x00007FF89A481A60+0xA2)
System.Web.dll!System.Web.UI.Page.ProcessRequest() (IL=0x0014, Native=0x00007FF89A4819B0+0x4B)
System.Web.dll!System.Web.UI.Page.ProcessRequest(System.Web.HttpContext context) (IL=epilog, Native=0x00007FF89A4809F0+0x69)
App_web_picker.aspx.9c9699a8.uaajqlnbx.dll!ASP._layouts_15_picker_aspx.ProcessRequest(System.Web.HttpContext context) (IL=0x0007, Native=0x00007FF89E050A50+0x2D)
System.Web.dll!System.Web.HttpApplication.CallHandlerExecutionStep.System.Web.HttpApplication.IExecutionStep.Execute() (IL=0x018D, Native=0x00007FF89A47FF20+0x335)
System.Web.dll!System.Web.HttpApplication.ExecuteStepImpl(System.Web.HttpApplication.IExecutionStep step) (IL=epilog, Native=0x00007FF8996CB5C0+0xD5)
System.Web.dll!System.Web.HttpApplication.ExecuteStep(System.Web.HttpApplication.IExecutionStep step, ref bool completedSynchronously) (IL=0x0015, Native=0x00007FF8996CB210+0x5B)
System.Web.dll!System.Web.HttpApplication.PipelineStepManager.ResumeSteps(System.Exception error) (IL=0x027A, Native=0x00007FF8996C8C20+0x74D)
System.Web.dll!System.Web.HttpApplication.BeginProcessRequestNotification(System.Web.HttpContext context, System.AsyncCallback cb) (IL=0x0031, Native=0x00007FF8996C8760+0x83)
System.Web.dll!System.Web.HttpRuntime.ProcessRequestNotificationPrivate(System.Web.Hosting.IIS7WorkerRequest wr, System.Web.HttpContext context) (IL=0x00B0, Native=0x00007FF8996C26E0+0x1E7)
System.Web.dll!System.Web.Hosting.PipelineRuntime.ProcessRequestNotificationHelper(System.IntPtr rootedObjectsPointer, System.IntPtr nativeRequestContext, System.IntPtr moduleData, int flags) (IL=0x0131, Native=0x00007FF8996C0830+0x3FD)
System.Web.dll!System.Web.Hosting.PipelineRuntime.ProcessRequestNotification(System.IntPtr rootedObjectsPointer, System.IntPtr nativeRequestContext, System.IntPtr moduleData, int flags) (IL=0x0000, Native=0x00007FF8996C03C0+0x13)
[Managed to Native Transition]
System.Web.dll!System.Web.Hosting.PipelineRuntime.ProcessRequestNotificationHelper(System.IntPtr rootedObjectsPointer, System.IntPtr nativeRequestContext, System.IntPtr moduleData, int flags) (IL=0x01E7, Native=0x00007FF8996C0830+0x4A4)
System.Web.dll!System.Web.Hosting.PipelineRuntime.ProcessRequestNotification(System.IntPtr rootedObjectsPointer, System.IntPtr nativeRequestContext, System.IntPtr moduleData, int flags) (IL=0x0000, Native=0x00007FF8996C03C0+0x13)
[Appdomain Transition]

```

And when the other `ItemPickerDialog` type is used, just the two topmost entries are different and then look like this:

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```
>
Microsoft.SharePoint.Portal.dll!Microsoft.Office.Server.ApplicationRegistry
.Infrastructure.EntityInstanceIdEncoder.DecodeEntityInstanceId(string encodedId)
(IL=0x0000, Native=0x00007FF89EA41610+0x23)
Microsoft.SharePoint.Portal.dll!Microsoft.SharePoint.Portal.WebControls.Item
Picker.ValidateEntity(Microsoft.SharePoint.WebControls.PickerEntity pe)
(IL=0x002C, Native=0x00007FF89EA40060+0x98)
```

This is the final proof that the data of

`ctl00$PlaceholderDialogBodySection$ctl05$hiddenSpanData` ends up in

`EntityInstanceIdEncoder.DecodeEntityInstanceId(string)`. The rest is only coping with entity instance id encoding and finding an appropriate `XmlSerializer` payload.

After the patch was made available in February, Markus noticed something unusual. The original patch only addressed the

`Microsoft.SharePoint.BusinessData.Infrastructure.EntityInstanceIdEncoder` in `Microsoft.SharePoint.dll` but not the

`Microsoft.Office.Server.ApplicationRegistry.Infrastructure.EntityInstanceIdEncoder` in `Microsoft.SharePoint.Portal.dll`.

By using the `EntityInstanceIdEncoder` type from the `Microsoft.SharePoint.Portal.dll` with the `Picker.aspx` as described here, the exploit still worked even though the patch was installed. Microsoft addressed this with the re-release of [CVE-2019-0604](#) yesterday.

Special thanks to Markus for providing us such a great write-up. Markus can be found on Twitter at [@mwulftange](#), and we certainly hope to see more submissions from him in the future. Until then, follow the [team](#) for the latest in exploit techniques and security patches.

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